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Reconstruction and analysis of pore blockage fouling in ceramic ultrafiltration membranes through FIB-SEM

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Pore blockage fouling significantly impairs ultrafiltration (UF) efficiency. Recent advances in volume electron microscopy have enabled nanoscale investigations of its mechanisms. Here, we developed a focused ion beam-scanning electron microscopy (FIB-SEM) tomography strategy that enables accurate 3D reconstruction of organic foulant accumulation within ceramic membrane pores. FIB-SEM achieves a voxel resolution of $2 \times 2 \times 5$ nm^3. The high-resolution imaging clearly reveals the trilayer structure of ceramic UF membranes, and captures the dynamic evolution of pore blockage fouling. The interface between the top and intermediate layers emerges as the primary site for initial membrane fouling. This localized channel constriction decreases the theoretical water permeability by 76.5%, directly explaining the rapid initial flux decline. Lattice Boltzmann simulations reveal that multilayer membrane structures create flow stagnation zones. Near-zero velocities trigger foulant accumulation, clarifying pore blockage mechanisms. 3D reconstruction and pressure distribution analysis revealed that pore blockage fouling is effectively removed during physical backwash, but could rapidly accumulate again at the initial stage of the next operation cycle. This study provides key insights into pore blockage fouling in trilayer ceramic UF membranes, and offers a theoretical basis for guiding the design of anti-fouling membranes.

Ultrafiltration (UF) has critical applications across water treatment fields, including in drinking water production systems[1], water reuse systems[2], desalination plants[3], and industrial separation processes[4]. Ceramic UF membranes show strong promise for these challenging applications because of their inherent chemical and thermal stability[5,6], particularly in decentralized continuous-operation scenarios such as rural drinking water treatment and seawater desalination[7]. To balance membrane flux and mechanical strength, composite membranes are typically fabricated by coating a selective top layer with small pores onto a macroporous support layer. UF membranes with a fine-pored top layer often include an intermediate layer between the support and the top layer. This intermediate layer enhances the surface flatness of the substrate during the coating process, thereby reducing the formation of defects in the top layer[5,8–10]. However, the wide implementation and long-term operation of ceramic membranes are greatly inhibited by membrane fouling, which leads to a decrease in the flux, an increase in the operating pressure, and membrane failure[11,12]. Generally, UF membrane fouling occurs in two forms: surface cake layer

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buildup and internal pore blockage[13]. Owing to the complex three-layer architecture, the pore blockage fouling mechanisms in ceramic membranes remain poorly understood, hindering effective fouling control. Therefore, a key challenge is that a nanometer-resolution map depicting the 3D pore network of ceramic UF membranes is lacking.

Ceramic UF membranes exhibit nanometer-scale pore structures and micrometer-thick separation layers[14]. Their 3D structural characterization requires cross-scale techniques. Inroads have been made with light microscopy and mass spectrometry, but the results are resolution limited. Light-sheet fluorescence microscopy (LSFM) achieves a resolution of only 3 μm[13], which is constrained by the Abbe diffraction limit[15]. The time-of-flight secondary ion mass spectrometry (ToF-SIMS)-based method achieves only a 5 μm in-plane resolution[16]. Researchers have utilized transmission electron microscopy (TEM)-based technologies to achieve 3D reconstructions of nanofiltration membranes with nanometer resolution[17–19]. However, these techniques require samples thinned to under 80 nm. The micrometer-scale membrane structures cannot be preserved in samples of such thickness.

Focused ion beam-scanning electron microscopy (FIB-SEM) is a cross-scale technique for 3D reconstruction[20]. Studies have shown that this technique creates 3D cell organelle reconstructions at a 4 nm resolution in all directions while handling structures up to the micrometer scale[21]. Additionally, notable variations in electron scattering properties among the ceramic matrix, organic foulants, and pore structures enable their distinct visualization and differentiation via SEM secondary electron detection. These characteristics make FIB-SEM a highly promising tool for 3D visualization of ceramic UF membrane architectures.

In this study, the pore blockage mechanisms in ceramic UF membranes was investigated. We acquired successive cross-sectional images of both clean and fouled ceramic UF membranes ($2 \times 2 \times 5\ nm^3$ resolution) through FIB-SEM. Simultaneously, automated image segmentation via deep learning was implemented, enabling 3D reconstruction of membrane substructures at the 2 μm scale. Pore network modeling and lattice Boltzmann simulations elucidated the microscale flow characteristics within the reconstructed architecture. This work reveals four critical findings: 1) the key factors determining the pore blockage severity, 2) the dynamic processes causing flux decline during membrane fouling, 3) the structural determinants driving fouling initiation, and 4) the reversibility of pore blockage during backwash. This study offers prospective guidance for optimizing UF membrane structures and designing anti-fouling ceramic membranes. The research approach can also be extended to investigate various porous media containing nanochannels.

Results

3D reconstruction framework for ceramic UF membranes

3D tomography based on FIB-SEM was employed to reconstruct the structure of ceramic UF membranes (CM15; the molecular weight cutoff specified by the manufacturer was 15 kDa) (Fig. 1a). To minimize ion beam-induced artifacts, platinum and carbon protective layers were first deposited on the membrane surface using a gas injection system (Supplementary Fig. S1). Sequential gallium ion (Ga^+) milling (30 kV, 300 pA) coupled with secondary electron imaging (2 kV, 500 pA) achieved a 2.0 nm/pixel resolution with a nominal slice thickness of 5.0 nm. Post-acquisition metrology revealed an actual average slice thickness of 5.2 nm (Fig. 1b), attributed to operational drift in ion beam control. The aligned slice series formed a clean volumetric dataset for subsequent computational analysis. Image segmentation was achieved through a deep learning approach[22]. Input datasets were generated through Otsu's method followed by manual refinements. The U-Net 2.5D architecture in the Intelligent Segmentation module completed model training, enabling accurate segmentation of pore structures

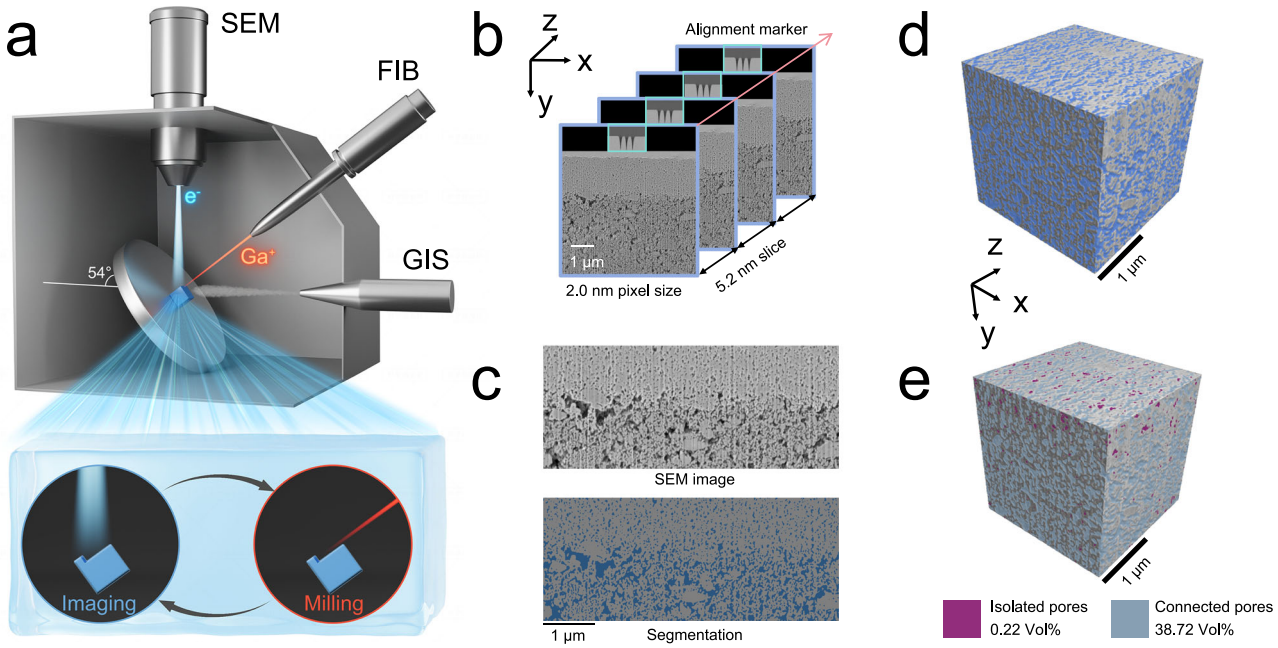


Fig. 1 | Schematic workflow of the FIB-SEM tomography-based reconstruction protocol for ceramic membranes. a Schematic illustrating the operational principle of FIB-SEM tomography. The process involves repeated cycles of milling the sample surface with a focused gallium (Ga^+) ion beam and imaging the newly exposed cross-section with an scanning electron beam. This sequential acquisition generates a stack of serial images for 3D reconstruction. The GIS is employed for depositing conductive protective coatings. **b** Registration and alignment of SEM images. This was achieved using pre-etched triangular markers on the platinum layer as reference points (highlighted by green squares). The images were acquired with a pixel size of 2 nm. The average thickness of the aligned slices is 5.2 nm. The red arrow indicates the milling direction. **c** Workflow for segmenting SEM grayscale cross-sectional images into pores and membrane components. The training dataset was generated by manually labelling pores (blue) and membrane matrix (gray). A deep learning model was trained on this dataset to perform the segmentation task automatically. **d** 3D reconstruction of a ceramic UF membrane, with the pores colored in blue. **e** Analysis of pore connectivity within the ceramic membrane shown in Fig. 1d. The connected pore network is depicted in blue, while isolated pores are colored in purple. Scale bar: **b**–**e** 1 μm.

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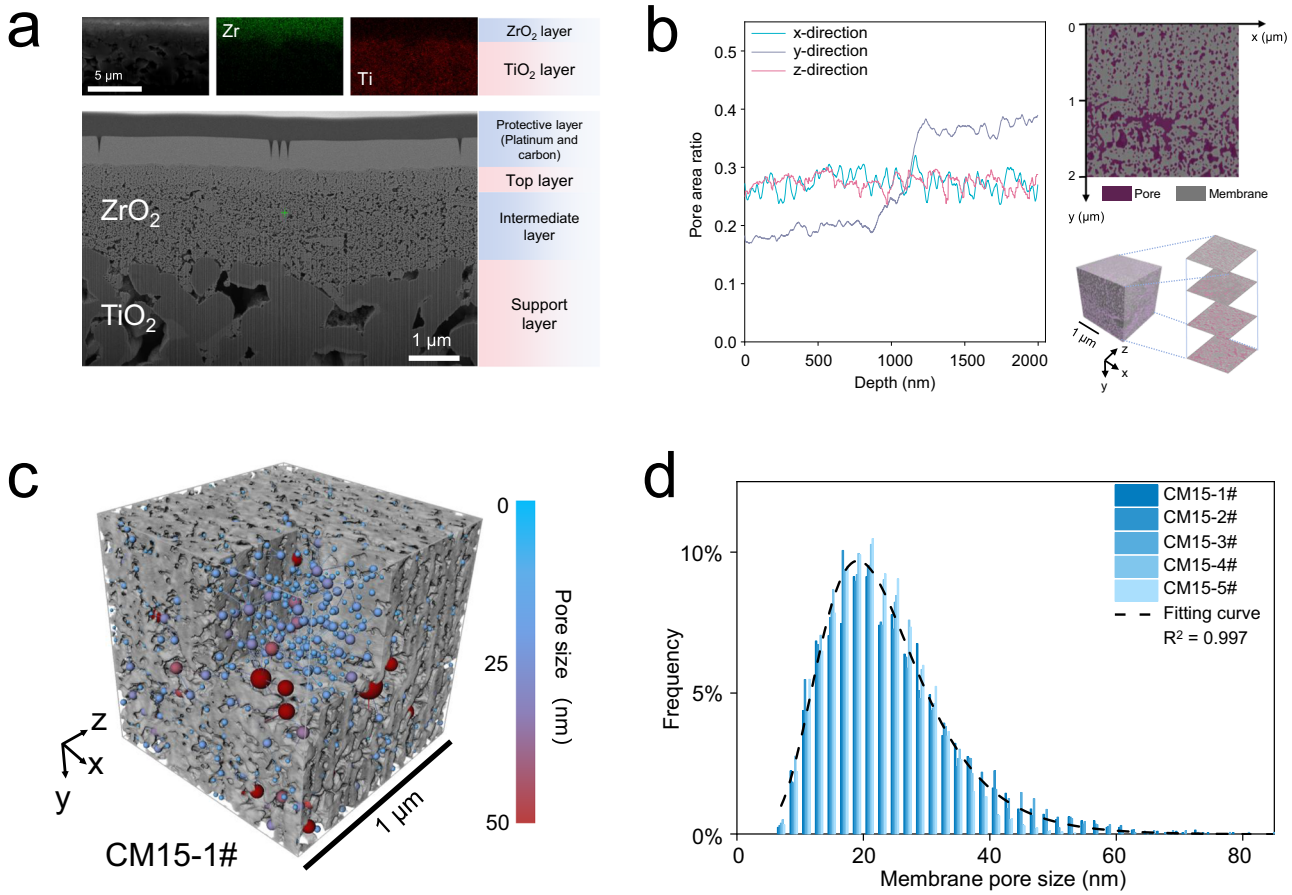


Fig. 2 | Structural parameterization of the ceramic membrane architecture. **a** SEM cross-section images of CM15, combined with EDS mapping of Zr (green) and Ti (red) elements. The image reveals the three-layer architecture and the corresponding material of each individual layer. **b** Variation of porosity with depth in the CM15 sample along the x, y, and z directions. The right panel displays the porosity characteristics on the xy-plane and illustrates the method for extracting the porosity profile along the y-direction. **c** Pore network model reconstructed from the binarized cross-sectional image. The color of the spheres represents the pore size, ranging from blue (small) to red (large). This model, designated as CM15-1#, is used to distinguish it from the other four models presented in the Supplementary Information. **d** Pore size distribution frequency for the five CM15 models, along with the corresponding right-skewed normal distribution fit to the data. Scale bar: (**a**–**c**) 1 μm.

within CM15 (Fig. 1c). The segmented model subsequently reached a uniform 2.0 nm resolution in all dimensions through upsampling, allowing comprehensive 3D reconstruction of the CM15 architecture (Fig. 1d). The pore connectivity in CM15 was further characterized. The top surface and bottom/lateral surface were demarcated via surface segmentation (Supplementary Fig. S2), and Boolean operations were applied to compute pore geometries intersecting both domains, thereby defining hydraulically connected pores. Quantitative analysis revealed that the connected pore network constituted 38.7% of the total analyzed volume, whereas the isolated pores accounted for as little as 0.22%. The results confirmed the superior pore connectivity of the clean CM15 (Fig. 1e).

Pore network model (PNM)-based 3D membrane analytics

The ceramic membrane cross-section was prepared using FIB milling and showed an asymmetric layered structure. Energy-dispersive X-ray spectroscopy (EDS) elemental mapping revealed that zirconium dioxide (ZrO_2) was present primarily in the top two layers and that titanium dioxide (TiO_2) was present mainly in the bottom sublayer (Fig. 2a) (Supplementary Text. 1). The TiO_2 supporting layer contained micrometer-sized pores formed by coarse particle packing, and this structure had no molecular sieving capability. The ZrO_2 layer consisted of two separate layers with a clear interface. The top layer had a dense microstructure with nanopores, whereas the contiguous intermediate layer had substantially enlarged pores. The porosity gradients across the spatial axes were measured using a representative $2 \times 2 \times 2\ \mu m^3$ volume covering both the top and intermediate layers (Fig. 2b). Isotropic slicing at 2.0 nm intervals produced 1000 cross-sections for porosity analysis. Along the *x*- and *z*-axes, a consistent porosity of approximately 0.25 was maintained. For the *y* axis, the porosity varied with depth. The porosity remained stable at approximately 0.18 in the top layer. Within the intermediate depth zone (870–1230 nm), the porosity progressively increased to 0.36 and then reached a plateau. Validation via independent CM15 tests confirmed the same pattern (Supplementary Fig. S3).

Five randomly selected CM15 ceramic membranes were employed for systematic investigation of the top layer structure. All the samples were subjected to standardized 3D reconstruction through the above established protocols. The pore network model generated with OpenPNM[23] quantified key structural descriptors (Fig. 2c and Supplementary Fig. S4). To validate the accuracy of the PNM, the CM15-1# sample was used as a representative model. A comparison of the permeability and tortuosity calculated by the PNM and the Lattice Boltzmann Method (LBM) was conducted (Supplementary Text. 2 and Text. 3). The results show that the permeability obtained from the PNM is 2.47×10^{-7} D, while that derived from LBM is 1.35×10^{-7} D, both values

falling within the same order of magnitude. The tortuosity calculated via the PNM is 1.76, compared to 1.87 from the LBM velocity field, yielding a relative error of only 5.9%. Therefore, from a hydraulic perspective, abstracting the ceramic UF membrane structure as a pore network model enables accurate quantitative analysis.

The pore size distributions across the five CM15 top layers were closely aligned, which confirmed their structural consistency (Fig. 2d). Gumbel distribution modeling effectively described the right-skewed dataset, with a central value of 19.0 nm (Supplementary Text. 4). The observed pore sizes ranged from 7.0 nm to 85.0 nm, with 95% of the sizes concentrated between 10.0 nm and 50.0 nm. This distribution model was applied to describe the structural characteristics of the CM15 top layer in follow-up research.

Membrane fouling properties

CM15 membrane fouling experiments were conducted using a cross-flow filtration system (Fig. 3a). All the experimental protocols are fully described in the Methods section. A transmembrane pressure of 1.0 bar was applied. Three model foulant categories, dextran (DEX), dextran–bovine serum albumin (DEX–BSA) composites, and polystyrene nanoparticles (PSNPs), were systematically selected to span broad size ranges while maintaining analogous physical properties. The nomenclature followed the particle characteristics: numerical suffixes denote the molecular weight (kDa) of dextran (e.g., DEX10 = 10 kDa dextran). Dynamic light scattering (DLS) analysis was used to determine the dextran particle diameter. The DEX10 to DEX2000 samples covered a size range of 1.5–28.8 nm. The diameter measurements linearly correlated with the cubic root of the molecular weight. This result corresponded with theoretical predictions[24] (Supplementary Fig. S5). Synthetic DEX–BSA exhibited a broad size distribution with peaks spanning from 32.7 to 342.0 nanometers (Supplementary Fig. S6a). The PSNPs, with manufacturer-specified size ranges of 50–100 nm (PSNP50) and 200–500 nm (PSNP200), were measured by TEM to have actual sizes of approximately 70 nm and 300 nm, respectively. (Supplementary Fig. S6b and S6c).

Figure 3b shows consistent fouling patterns across all experimental groups. The flux-time curve indicates a rapid flux decline during the initial 4 min, followed by a more gradual reduction. Based on the Hermia model[25–27], fouling evolution can be inferred from membrane flux decline profiles as detailed in Supplementary Text. 5. Taking DEX800 fouling as an example (Supplementary Fig. S7), the dominant fouling mechanism shifts in two distinct phases: During the initial 4 min, combined complete blocking and standard blocking mechanisms prevail. Following a transition phase, cake layer formation dominates. Thus, the acute flux decline observed within the initial 4 min is primarily attributed to pore blockage fouling. This is corroborated by SEM imaging of the CM15 membrane surface at the 4-min mark (Supplementary Fig. S8), which shows no significant cake layer formation, further confirming that the initial flux decline was mainly caused by pore blockage.

After 4 min of fouling, PSNP200 caused the smallest flux reduction (12.8%), whereas DEX2000 produced the largest flux decline (82.8%). From 4 to 30 min, all the groups displayed similar flux reductions (5.0–12.9%) (Fig. 3c). These results demonstrate that pore blockage fouling occurring during the initial filtration stage significantly contributes to the decrease in the membrane flux, with the magnitude of this effect being determined by the foulant particle size.

As shown in Fig. 3d, the severity of membrane fouling is strongly correlated with the size of the foulant particles relative to the membrane pores. DEX10 (1.5 nm) and DEX40 (6.7 nm), with particle sizes below the pore size range of CM15, exhibited a limited capacity for complete pore obstruction. The flux decreases within 30 min were 34.8% and 49.3%, respectively. Their incomplete blocking behavior resulted in a relatively mild flux decline. The DEX150 (10.5 nm) particles matched the CM15 membrane pores, leading to direct blockage of the pores. Blocked pores formed dead-end structures, promoting foulant accumulation. This triggered an 81.1% flux decline within 30 min. DEX800 (20.2 nm) and DEX2000 (28.8 nm) intensified this trend, reaching 88.0% and 89.1% flux losses, respectively. The DEX–BSA sizes (32.7–342.0 nm) partially overlapped with those of the CM15 pores (7.0–85.0 nm). Size exclusion prevented most of the foulants from entering the pores. Consequently, the flux decline decreased to 58.4%. Both PSNPs were too large to cause internal membrane blockage fouling, resulting in flux declines of 28.3% for PSNP50 and 20.1% for PSNP200. FIB-SEM cross-section imaging verified the pore blockage patterns (Fig. 3e). After 30 min, DEX10-induced pore blockage was negligible in the CM15 membranes, whereas DEX150 and DEX800 were distinctly deposited within the pores. No internal blockage occurred with PSNP200 exposure. Thus, pore blockage requires contaminant–membrane pore size matching.

Furthermore, we validated this trend using natural organic matter (NOM), which is a common mixture of organic compounds in water and a significant contributor to UF membrane fouling. Experiments were conducted with SRNOM 2R101N at a concentration of 2 g·L^{-1}. The results indicate that the flux decline induced by NOM falls between that caused by DEX40 and DEX150 (Supplementary Fig. S9a). This intermediate behavior can be attributed to the abundance of macromolecular humic acids in NOM. Similar to the DEX–BSA system, NOM contains fewer components capable of pore blockage. Supporting evidence is provided by 3D reconstruction of membranes after 2 min of fouling (Supplementary Fig. S9b), which clearly demonstrates that NOM results in comparatively less obstruction of membrane pores.

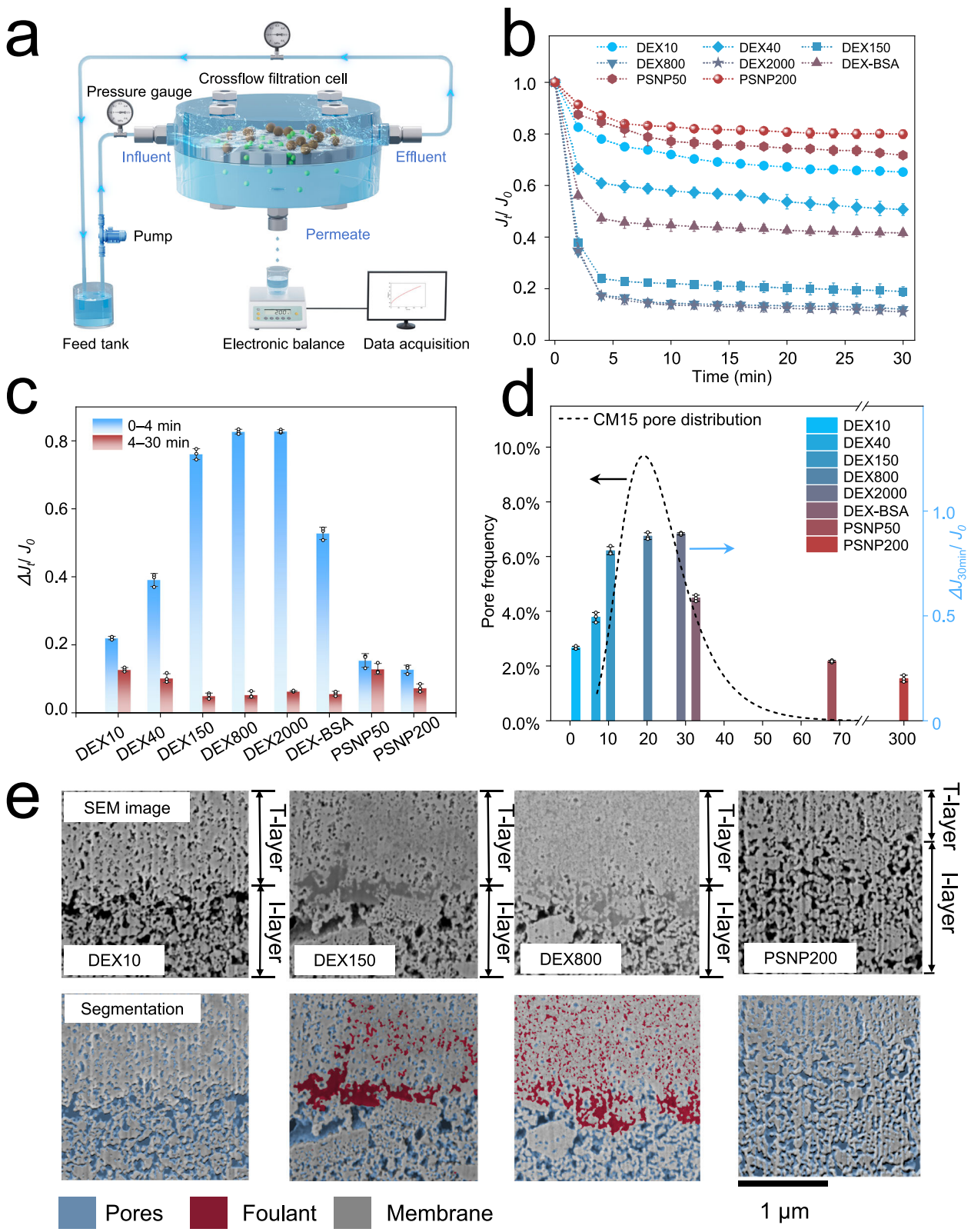
Pore blockage formation process and flux decline mechanisms

In this study, the pore blockage formation process in ceramic membranes was investigated using DEX800 as a model foulant because the diameter of DEX800 exactly overlaps with the dominant pore size of CM15, and a rapid decrease in the flux was observed after 4 min of DEX800 exposure ($\Delta J/J_0 > 80\%$). FIB-SEM nanotomography causes permanent membrane damage, thus requiring separate UF membranes for each 3D reconstruction. Nevertheless, batch uniformity tests on CM15 membranes confirmed highly consistent structural features (Fig. 2d). These structural differences demonstrably do not affect the fouling behavior (Fig. 3b).

Three fouling stages (0, 2, 4 and 30 min) were systematically analyzed to track structural changes during membrane fouling (Fig. 4a). Cross-sectional imaging revealed distinct deposition dynamics. Initially, foulants adhered solely to the interface between the top and intermediate layers (2 min), followed by outward migration toward the membrane surface (4 min). After 30 min, the pores in the top layer approached near-complete blockage. The 3D reconstructions provide a more spatially resolved visualization of this trend (Fig. 4b). DEX150 and DEX2000 were employed as model substances to experimentally verify this phenomenon, with consistent results (Supplementary Fig. S10).

A consensus has not yet been reached regarding the initial site of membrane fouling. While the general view posits that fouling begins at the membrane surface before gradually penetrating the pores[28,29], a few studies suggest it may begin internally within the membrane[30,31]. Given that different filtration methods (e.g., constant flux vs. constant pressure, cross-flow vs. dead-end) are used across the literature, we investigated the dead-end, constant-flux mode to help reconcile these conflicting viewpoints (Supplementary Fig. S11a). DEX800 was employed, acting as a model foulant. The flow rate was set to 60 L m^{-2} h^{-1} using a peristaltic pump. The results show that flux declined slower in constant-flux filtration than in constant-pressure filtration (Supplementary Fig. S11b). Under constant-flux filtration, pressure rose continuously from 0.40 bar to 2.05 bar at 10 min, finally stabilizing around 2.40 bar(Supplementary Fig. S11c). Two time points were chosen for structure analysis (2 min and

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10 min). The 2-min point reflects the early rapid flux drop, while the 10-min point marks the start of flux stabilization.

In dead-end constant-flux filtration, a cake layer formed quickly within 2 min (Supplementary Fig. S11d). By then, flux had fallen to 55.7% of the initial value, with no clear pore blockage yet (Supplementary Fig. S11e and S11f). This confirms that cake layer fouling alone can cause significant flux loss. After 10 min of fouling, the membrane surface was completely covered by the cake layer (Supplementary Fig. S11g), and internal pore blockage began to occur (Supplementary Fig. S11h and S11i). Notably, under constant-flow filtration mode, pore blockage fouling still initially appeared at the interface between the top layer and the intermediate layer.

Based on the above findings, it can be concluded that both cake layer fouling and membrane pore blockage are capable of causing a

Fig. 3 | Fouling characteristics of the CM15 membrane by organic foulants of different sizes. a System configuration for cross-flow UF experiments with integrated flux monitoring. **b** Normalized flux decline over time for the CM15, which showed the effect of foulant hydrodynamic size on membrane fouling. **c** Comparison of flux decline percentages during the initial (<4 min) and mid-late stages (>4 min) of the UF process. **d** Hydrodynamic size distribution of the foulants relative to CM15 membrane pore dimensions. The X-axis denotes the particle diameters for the dextran species and PSNPs. The minimal diameter is plotted for the DEX−BSA aggregate. The left Y-axis represents the pore size frequency of the CM15 membrane (black curve, fitted from Fig. 2d), while the right Y-axis shows the resultant flux reduction ratio (bar graph) for each foulant. **e** Cross-sectional SEM characterization and segmentation of membrane fouling induced by dextran species and PSNP200. The top row shows the original SEM images, with approximate regions of the top layer (T-layer) and the intermediate layer (I-layer) indicated. The bottom row presents false-colored images after segmentation to enhance visualization, where red areas denote foulants and blue areas represent unpolluted membrane pores. The concentration of the solutions were 2 g L^{-1}. Scale bar: **e** 1 μm. Error bar: (**b**–**d**) Standard Deviation (SD) of triplicates.

substantial reduction in flux, and the sequence of their occurrence is closely related to the UF mode. However, regardless of the filtration mode employed, membrane pore blockage consistently initiates at the interface between the top layer and the intermediate layer.

PNMs of CM15 under 0, 2, 4, and 30 min fouling conditions were reconstructed using OpenPNM for quantitative analysis (Fig. 4c). The clean membrane demonstrated a relatively homogeneous porous structure in the top layer, which was indicated by uniform blue spherical cluster distributions. The intermediate layer showed localized pore expansion features marked by red clusters. After 2-min fouling, severe pore blockage was observed at the interface between the top and intermediate layers. A 52.1% pore blockage ratio was recorded in critical fouling zones (Supplementary Fig. S12a). A limited cluster of membrane pores remained active as hydraulic bottlenecks. After 4 min of fouling, the pore blockage had extended upward into the top layer, causing partial blocking. By the 30-min mark, the top layer was nearly entirely clogged, exhibiting a local pore blockage volume ratio of 66.0% (Supplementary Fig. S12b). The water transport pathways within this layer were reduced to a handful of remaining channels.

The theoretical permeability for the four fouling scenarios were calculated based on the pore network model. Supplementary Text. 6 contains the full computational parameters and theories. Supplementary Fig. S13 presents the calculation results. Compared to the permeability of the unfouled membrane structure, the theoretical permeability decreased to 38.1%, 15.7%, and 12.6% at 2, 4, and 30 min, respectively. The theoretical calculations involve inherent uncertainties, and the selected micrometric samples might not fully represent the entire membrane. Nevertheless, the calculated permeation patterns were highly consistent with the experimental flux reduction trends. This suggests that the structural alterations resulting from local pore blockage help to explain the overall reduction in flux.

The fouled membrane model can be divided into three distinct zones: the upper bottleneck zone (Region #1), primary bottleneck zone (Region #2), and lower bottleneck zone (Region #3) (Supplementary Fig. S14a). Under the framework of Darcy's law, the total permeability governing the flow through three serially connected zones can be mathematically formulated as (Supplementary Text. 7):

$$k_{total} = \frac{L_1 + L_2 + L_3}{\frac{L_1}{k_1} + \frac{L_2}{k_2} + \frac{L_3}{k_3}} \quad (1)$$

When all three regions are roughly the same size (Fig. 2b), the area with the strongest flow resistance controls the overall permeability. The initial 2-min fouling resulted in significant degradation of the flow pathways in Region #2, leading to a sharp decline in k_2 and a rapid collapse of k_{total}. Subsequent fouling in Region #1 caused a notable reduction in k_1, and k_{total} exhibited a persistent decline. Similarly, the significant flux reduction in constant-flux dead-end filtration is primarily attributed to the rapid development of a cake layer formation in the initial fouling stage.

Pore blockage formation mechanism

Figure 2b reveals measurable porosity enlargement between the top layer and intermediate layer. This structural configuration likely initiates contaminant deposition. Quantitative membrane pore analysis confirmed persistent size variations across depth. The CM15 membrane displays three vertically stratified regions on the basis of the classification criteria established in the previous section. Two-dimensional cross-sectional analysis revealed progressive pore enlargement with increasing depth (Supplementary Fig. S14a). Pore network modeling across three membrane regions clarified this size progression through systematic distribution analysis (Supplementary Fig. S14b). Notably, the interface between the top and intermediate layers manifests dual structural adjustments: individual pore diameter expansion accompanies a collective porosity increase. These coupled modifications inevitably alter the local flow characteristics.

The LBM was applied to analyze the flow behavior in the ceramic membranes. Dimensionless parameters were used in these calculations. The Methods section fully describes the computational framework and boundary conditions. After 2 min of fouling, foulants concentrated at the interface between the top and intermediate layers and were located approximately 1 μm below the surface (Fig. 5a). LBM-generated velocity fields detected initial flow deceleration at the interface between the top and intermediate layers. Both flow monitoring (Fig. 5b) and cross-sectional evaluation (Fig. 5c) verified this phenomenon. The mean flow rates across depth-specific cross-sections were used to quantify the velocity distribution (Fig. 5d). The cross-sectional mean flow velocity was 8.07×10^{-5} at a depth of 0.176 μm, which progressively decreased to 2.87×10^{-5} at a depth of 1 μm, only 35.6% of the initial value. Furthermore, the velocity contour revealed multiple deep blue regions in cross-sectional views (Supplementary Fig. S15). These areas demonstrated localized flow stagnation, where the flow velocity approached zero. Reduced flow rates facilitated pollutant accumulation within membrane pores. Once pore blockage occurred, these passages became dead ends. The subsequent foulants could not permeate through them and continued to accumulate, ultimately propagating upward to form near-complete blockage in the top layer.

The impact of backwash on pore blockage fouling

The characteristics of membrane pore blockage fouling during the backwash process were analyzed (Fig. 6). The experiment used DEX800 as a model pollutant to foul CM15 for 30 min, followed immediately by a 5-min physical backwash, with an operating pressure of 1.0 bar. This cycle was repeated 10 times. Detailed experimental procedures are provided in the Methods section. The results indicate that physical backwash cannot completely remove membrane fouling. After the first cycle, the membrane flux decreased to 49.8% of the initial flux. As the number of cycles increased, the degree of flux recovery gradually declined. After ten cycles, the post-backwash membrane flux was only 29.3% of the initial flux (Fig. 6a). In contrast, chemical cleaning effectively restored the membrane flux.

The CM15 membrane after the tenth physical backwash cycle was selected for analysis. During the fouling-backwash cycles, the fouling on the surface of CM15 was not completely removed, with noticeable pollutants still adhering to it (Fig. 6b). Cross-sectional electron microscopy images revealed that the residual fouling was mainly concentrated on the surface, while the internal membrane pore blockage was almost entirely eliminated (Fig. 6c).

A 3D reconstruction of a localized area of CM15 under backwash conditions showed that the backwash water flows from the intermediate

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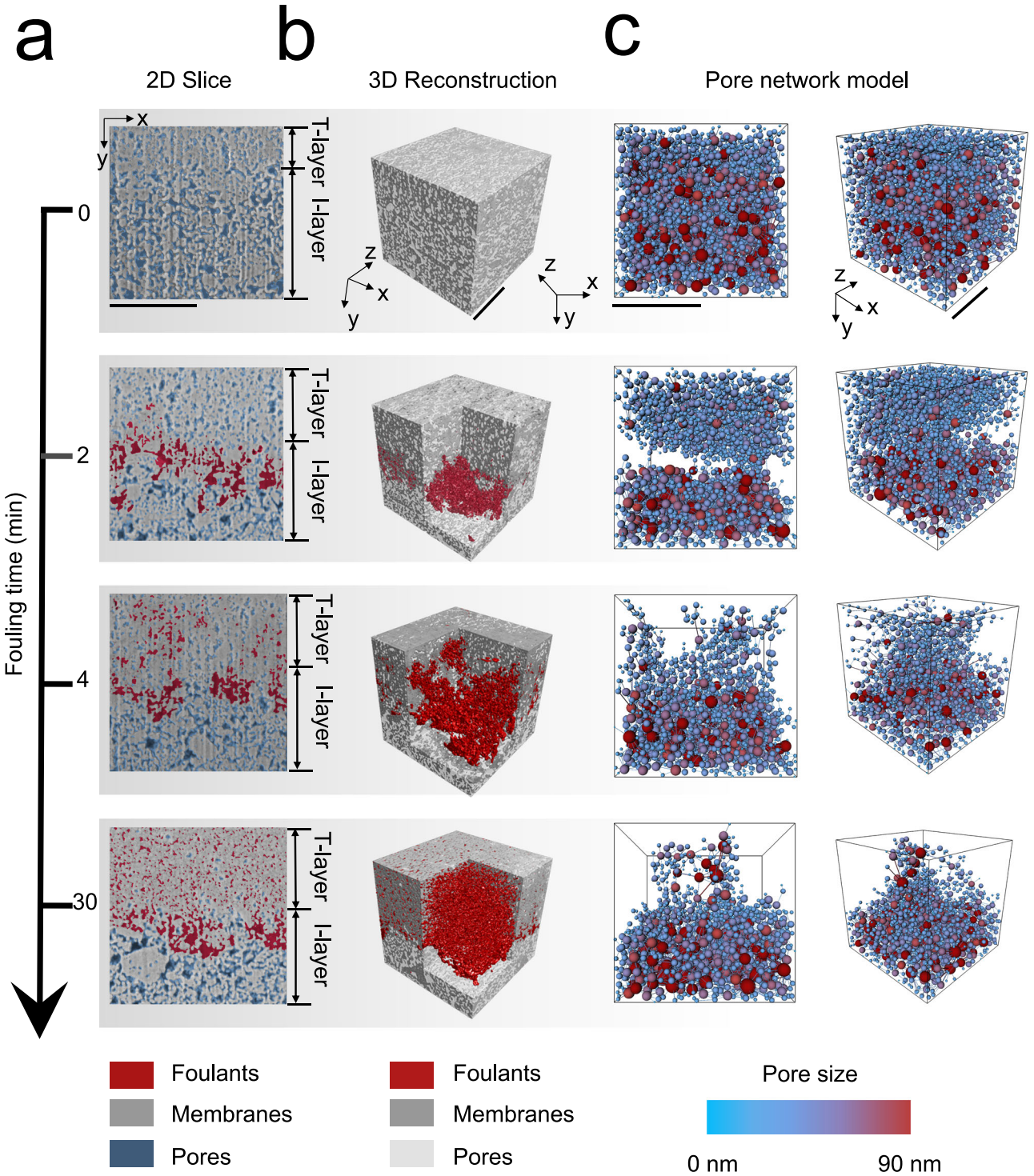


Fig. 4 | The fouling process (0 min, 2 min, 4 min and 30 min) of DEX800 on CM15 revealed by 3D reconstruction and pore network modeling of water-transport channels. a 2D cross-sections of the CM15 membrane at different fouling times (0, 2, 4, and 30 min), showing the spatial distribution of foulant. The original SEM image was pseudo-colored to highlight fouled pores (red) and unpolluted pores (blue). The approximate regions of the T-layer and the I-layer are indicated. **b** 3D reconstruction of membranes corresponding to the time points in visualizing the fouling process. The foulants are colored in red. The images illustrate that fouling initiates at the top layer and intermediate layer interface and propagates upwards, eventually blocking the top layer. **c** PNMs of the water-transport channels in membranes at different fouling times. Each sphere in the PNM represents a pore, colored according to its size from small (blue) to large (red). The models show a progressive reduction in the density of water-transport channels within the top layer as fouling time increases. The concentration of DEX800 was 2 g L^{-1}. Scale bar: (**a**–**c**) 1 μm.

layer to the top layer (Fig. 6d). Based on the 3D reconstruction, a pore network model was established (Fig. 6e) to calculate the pressure distribution (Fig. 6f) and determine the average pressure across different depth intervals (Fig. 6g). The results demonstrated that during backwash, the membrane interior maintains relatively high pressure. Even at a distance of 200 nm from the membrane surface, the backwash pressure remains above 0.4 bar. This is likely due to the small pore size and severe blockage in the top layer, where the pressure drop is concentrated.

Thus, backwash exerts high pressure and impact on internal pore blockage fouling, facilitating its removal. However, the membrane

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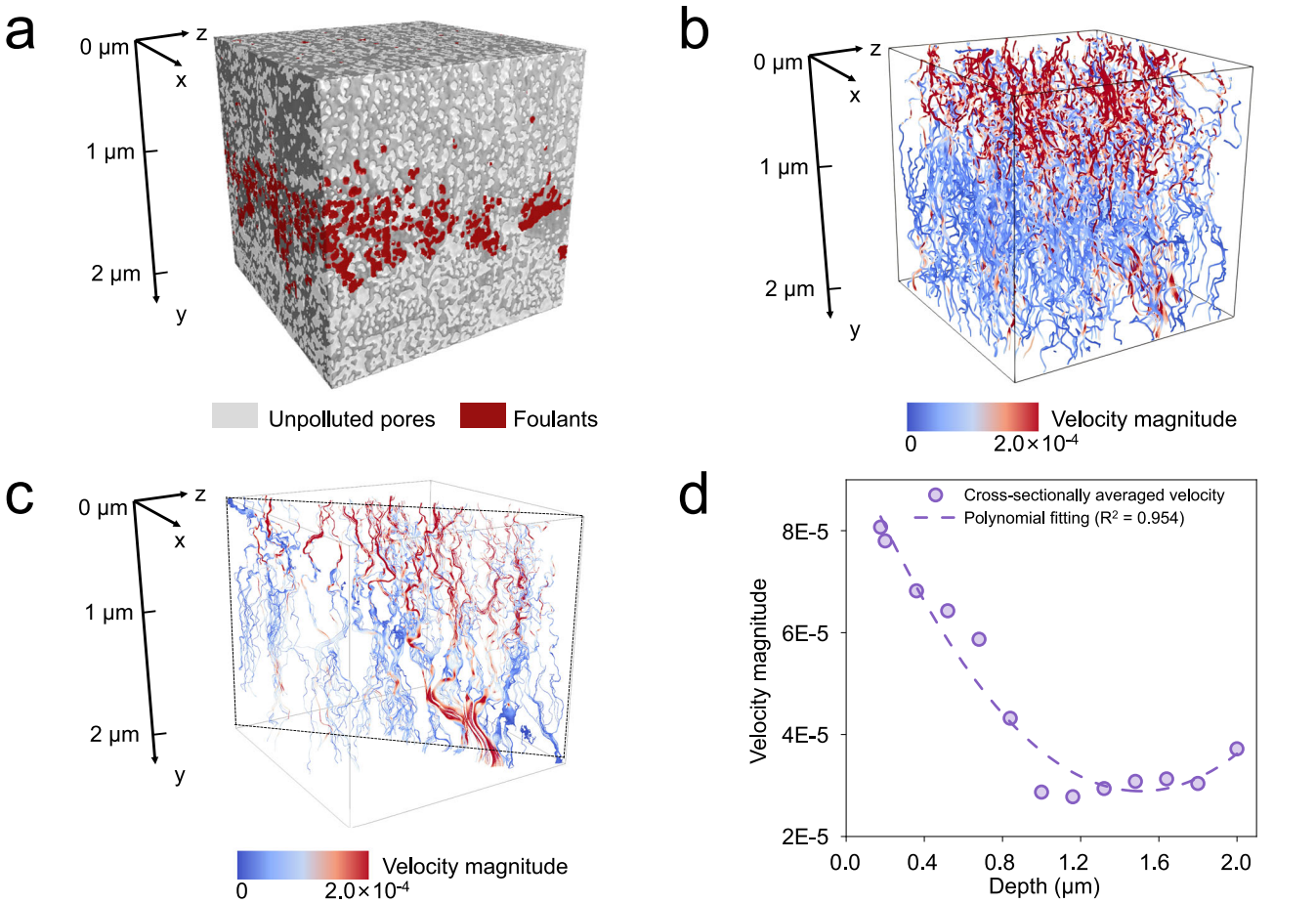


Fig. 5 | Hydrodynamic analysis of the solution transport across the membrane during the initial stage of ultrafiltration. a 3D reconstruction of the CM15 after 2 min of fouling by 2 g L^{-1} DEX800. Foulants are colored in red. **b** Perspective view and (**c**) cross-section of the water diffusion pathways through the clean CM15. The structure of the clean membrane is derived from the data in Fig. 5a by virtually replacing the foulants with unpolluted membrane pores. The streamlines, colored according to the corresponding flow velocity, are shown. This velocity is a dimensionless quantity derived from LBM simulations, with the color scale ranging from blue (low) to red (high). **d** Cross-sectionally averaged velocity in the sections at different depths. The velocities were calculated based on the simulation results shown in Fig. 5b.

surface is exposed to the atmosphere during backwash, resulting in a relative pressure close to zero. Additionally, the backwash water experiences a significant pressure drop after passing through the top layer, leading to limited removal efficiency for surface fouling. Therefore, while physical backwash is highly effective at removing internal pore blockage, the primary residual fouling remains on the membrane surface.

As shown in Fig. 6a, although backwash effectively removes pore blockage fouling, such fouling rapidly reoccurs early in the subsequent filtration cycle. The incompletely cleaned membrane surface and the reoccurrence of pore blockage significantly restrict membrane flux. In this study, CM15 operated below 16% of its initial flux for more than 86% of the running time. Hence, simultaneous removal of cake layer fouling and pore blockage fouling is crucial for ensuring the efficient operation of UF.

Discussion

We achieved 3D reconstruction of ceramic UF membranes at a 2 × 2 × 5 nm^3 resolution using FIB-SEM and deep-learning-based image segmentation. In this work, pore structure variations in asymmetric ceramic UF membranes during organic fouling were comprehensively visualized. The severity of internal pore blockage correlates directly with the foulant particle size: the most severe fouling occurs when foulant sizes fall within the membrane pore distribution range. The abrupt pore enlargement from the top layer to the intermediate layer causes a localized flow velocity reduction, leading to foulant accumulation. This accumulation reduces the effective flow area, creating a bottleneck effect that drives a rapid flux decline during initial filtration (e.g., 83.3% maximal flux drop at 4 min). This study, from a 3D structural perspective, confirms that membrane pore blockage is more likely to form during the initial stage of UF in a cross-flow system. This type of fouling is a major contributor to the significant flux decline that occurs initially. However, during the physical backwash process, the pressure gradient distribution makes pore blockage easier to remove, whereas the cake layer is more difficult to eliminate. Both cake layer fouling and pore blockage can cause a substantial decrease in membrane flux. Therefore, future research on anti-fouling UF membranes must consider strategies to remove both types of fouling. For example, modifying membrane surface properties such as hydrophilicity and electrical charge can reduce pollutant adhesion and suppress cake layer formation. Meanwhile, depositing catalytic materials at the interface between the top layer and the intermediate layer could allow reactive species to diffuse within a confined space. This would help suppress pore blockage at this critical location. This approach aims to remove fouling from key areas with low energy input, while maintaining the physical sieving function of the ceramic UF membrane. It avoids the need to make the entire membrane catalytic, which would compromise its sieving ability and turn it into a penetrative electrode.

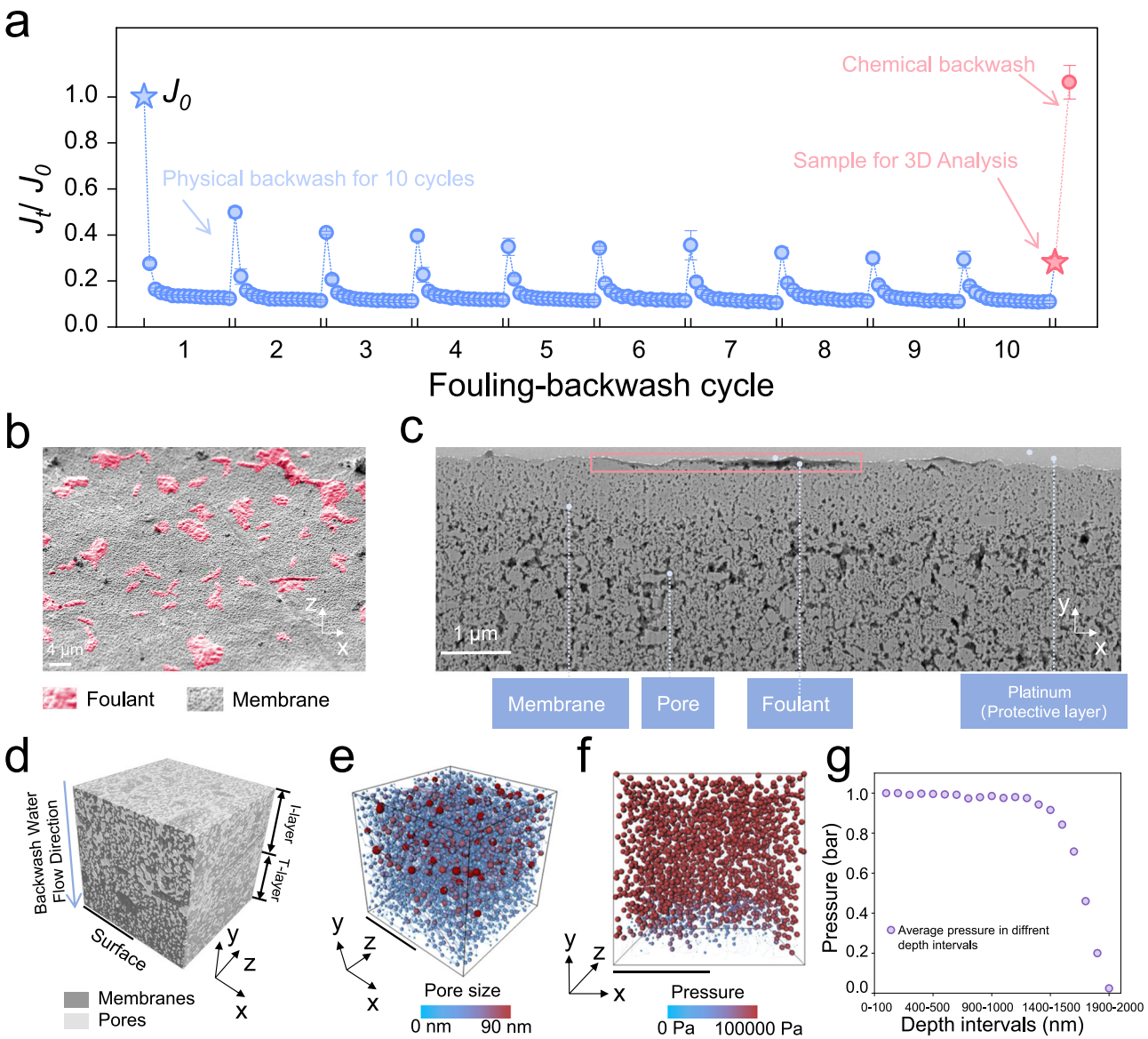


Fig. 6 | Characteristics of membrane fouling during backwash. a Normalized membrane flux as a function of time over ten repeated fouling-physical backwash cycles, followed by one chemical backwash with a 2 wt% NaOH solution. All flux values are normalized relative to the initial water flux. The membrane was fouled using a 2 g L^{-1} DEX800 solution. The membrane sample was collected after the final physical backwash for the subsequent analysis. **b** Surface SEM image of the CM15 membrane after ten fouling-physical backwash cycles. The foulants on the surface are colored red for enhanced visualization. **c** Cross-sectional SEM image of the CM15 membrane after ten fouling-physical backwash cycles. The red box highlights the fouling layer on the membrane surface from the cross-sectional perspective. **d** 3D reconstruction of the CM15 during backwash. The membrane is oriented surface-down, with the backwash flow direction from the I-layer to the T-layer. **e** Pore network model derived from the 3D structure in Fig. 6d. Spheres represent pores, colored according to their size from blue (small) to red (large). **f** Pressure distribution within the pore network during backwash. Spheres represent pores, colored by their local pressure from blue (low) to red (high). **g** Average pressure of pores as a function of depth. Data are binned every 100 nm, and the average pressure within each bin is plotted. Depth zero is defined at 2000 nm inside the membrane, with increasing depth corresponding to proximity to the membrane surface, aligned with the backwash flow direction. Scale bar: (**b**) 4 μm, (**c**–**f**) 1 μm. Error bar: (**a**) SD of triplicates.

Methods

Materials

Dextran with a molecular weight range of 10,000–2,000,000 g mol^{-1} for ultrafiltration studies was sourced from Yeasen Biotechnology (Shanghai, China) and demonstrated ≥99% chemical purity. High-molecular-weight dextran could not be obtained commercially. An alternative fabrication method with established protocols was employed to synthesize dextran-coated BSA particles (> 32.7 nm diameter), as detailed in prior literature[32] (Supplementary Text. 7). No polysaccharide-based material matching the 200 nm size requirement was available. PSNPs (200–500 nm) served as surrogate foulants for these trials. Given that these dimensions exceeded the ceramic membrane pore sizes (Fig. 3c), these particles were intentionally selected to demonstrate that contaminants incapable of entering the membrane pores cannot cause pore blockage fouling. While PSNPs represent a nonideal substitute under material constraints, their use to isolate and verify this physical exclusion mechanism remains methodologically justified. This research specifically investigates how pore blockage fouling is influenced by the relative size of pollutants and pores. To target this interaction, we chose the CM15 because its pore size distribution aligns closely with the size range of the dextran molecules.

Membrane fouling and backwash experiment

The presence of surface adsorbates significantly impacts the interfacial hydrophilicity of CM15, necessitating robust surface cleansing protocols to establish predictable hydrophilic pore characteristics for

subsequent permeability analyses. The CM15 test samples were secured on perforated alloy substrates inside a plasma reaction vessel in a BSET NT3 plasma reactor system (Glentech Inc., United States). Prior to hydrodynamic evaluation, the membranes were subjected to atmospheric plasma exposure at 18 W RF power for 20-min cycles to achieve optimal surface energetics. All foulant samples were uniformly dissolved in phosphate-buffered saline (PBS, pH 7.4) supplemented with 0.5% (w/v) Tween® 80 to inhibit particle agglomeration and ensure solution homogeneity. In replicate experiments, we employed multiple CM15 membranes from the same manufacturing batch to assess whether inherent structural variations exert nonnegligible impacts on the fouling behavior.

Preceding flux characterization trials, 10 liters of deionized (DI) water underwent cross-flow circulation through the ultrafiltration membranes for 24-h stabilization conditioning. The DI water was subsequently replaced with test analytes—aqueous dextran suspensions of distinct molecular sizes. A constant transmembrane pressure ($\Delta P = 1.0$ bar system average) was maintained during the filtration measurements. The cross-flow flow rate of the ultrafiltration system was maintained at 30 L h^{-1}. Based on the flow cross-sectional area in the membrane module, the cross-flow velocity is estimated to be approximately 0.057–0.092 m s^{-1}. The cross-flow flow field on the membrane surface was modeled as a cylinder with a base area of 30 mm^2 and a height of approximately 2 mm. By simplifying the cross-flow as a uniform flow, the velocity at the maximum cross-sectional area was estimated to be approximately 0.14 m s^{-1}. Permeate effluents were collected in gravimetric vessels positioned on computerized analytical balances (0.01 g resolution), with the mass acquisition intervals synchronized to 5-sec increments. The membrane flux can be calculated using the following formula:

$$J_i = \frac{(w_{i+1} - w_i)}{\rho_{water} \times A \times T \times \Delta P} \quad (2)$$

J_i represents the system flux at the i-th measurement, w_i is the mass reading from the i-th measurement, A denotes the effective membrane area, and T refers to the time interval. Continuous tracking of the flux decline under dynamic flow variations elucidates the progression of membrane fouling.

Backwash was performed using the same setup, with the key difference that the UF membrane was reversed. It was positioned with the support layer facing the incoming flow and the top layer facing the outflow. The backwash used DI water and the pressure was set at 1.0 bar. In each fouling-backwash cycle, the membrane was fouled for 30 min, followed by a 5-min backwash. Chemical backwash follows the same procedure as physical backwash, with the exception of using a 2 wt% NaOH solution instead of DI water. All experiments involving the measurement of membrane flux changes were performed three times, with the results averaged.

Focused ion beam scanning electron microscopy

In a FIB-SEM, an ion beam was employed for milling the UF membrane, whereas an electron beam was utilized for imaging both the surface and milled cross-sections. Prior to being loaded into the microscope chamber, the UF membrane was secured on the sample stage using a specialized adhesive. Owing to the nonconductive nature of the membrane itself, a high-angle Pt sputtering technique was implemented to deposit an approximately 3 nm-thick metallic layer on the membrane surface. This approach effectively enhances the surface conductivity while minimizing potential alterations to the native surface morphology of the UF membrane caused by the metal coating.

Following insertion into the microscope chamber, the sample stage was tilted at 54° to achieve orthogonal alignment with the ion source. Ga^+ ions accelerated at 30 kV with a beam current of 300 pA were employed to sequentially mill the ceramic membrane in 5-nm-thick slices per iteration. Post-milling operations utilized an electron beam operating at an accelerating voltage of 2 kV and a 500 pA beam current, achieving a resolution of 2 nm. The imaging area was strategically selected on the basis of real-time observations of the milled cross-section, with priority given to comprehensive coverage of the separation layer throughout the iterative cutting–imaging sequence. Following the image acquisition process, the total cutting depth is calculated from the focal distances at the initial and final positions, and the average thickness of each slice is then determined.

Image segmentation for the UF membrane, membrane pores, and contaminants was performed using deep learning-based method. The workflow proceeded as follows: First, the Otsu's thresholding algorithm was implemented to preliminarily delineate the membrane boundaries. Second, ten representative images were selected for manual refinement of threshold segmentation results, with distinct classifications established for the membrane matrix, pores, and foulants to generate input parameters for deep learning model training. Third, the U-Net architecture was subsequently used to process manually segmented slice data, and a 2.5D framework with 3 adjacent integrated slices configured with a depth level of 5 and an initial filter count of 64 was employed for deep learning training. Finally, the trained deep learning model processed all the raw datasets to generate 3D reconstructions.

Pore-scale numerical simulation method

Two methodologies were primarily employed for the numerical simulations at the pore scale: the PNM and the LBM. To analyze the membrane pore morphology and distribution characteristics, the PNM was adopted because of its advantages of lower computational demands and intuitive graphical representation. Conversely, to elucidate the formation mechanisms of membrane pore fouling, for which high-precision computational results are essential, the LBM was implemented because of its superior resolution capabilities. The PNM was constructed on the basis of the watershed algorithm, with a segmentation neighborhood of 26 and a marker extent of 1. In the LBM framework, the D3Q19 lattice configuration was employed, enabling comprehensive analysis and precise quantification of spatially resolved pressure and velocity profiles within critical membrane structural domains through systematic numerical resolution[33]. The pressure boundary condition was imposed at the inlet and outlet, whereas the no-slip bounce-back scheme was adopted for the solid boundaries. Dimensionless numerical parameters were employed in the LBM calculations. A pressure of 1.5×10^{-4} was maintained at the inlet boundary, whereas the outlet boundary operated under zero-pressure conditions. The working fluid was assigned a dynamic viscosity of 0.001 Pa·s, ensuring physiochemical consistency with water-based filtration systems. The detailed calculation theories and steps are given in the Supplementary Text 8.

Data availability

The authors declare that the main data supporting the findings of this study are available within the article and its Supplementary Information files. Extra data are available from the corresponding author upon request. Source data are provided with this paper.

References

1. Werber, J. R., Osuji, C. O. & Elimelech, M. Materials for next-generation desalination and water purification membranes. *Nat. Rev. Mater.* **1**, 1–15 (2016).
2. Yang, J. et al. Cobalt single-atom catalyst tailored ceramic membrane for selective removal of emerging organic contaminants. *Environ. Sci. Ecotechnol.* **21**, 100416 (2024).
3. Ranieri, L., Esposito, R., Nunes, S. P., Vrouwenvelder, J. S. & Fortunato, L. Biofilm rigidity, mechanics and composition in seawater

desalination pretreatment employing ultrafiltration and microfiltration membranes. *Water Res.* **253**, 121282 (2024).
4. Zhang, H. et al. Ultrafiltration separation of am(VI)-polyoxometalate from lanthanides. *Nature* **616**, 482–487 (2023).
5. Li, C., Sun, W., Lu, Z., Ao, X. & Li, S. Ceramic nanocomposite membranes and membrane fouling: A review. *Water Res.* **175**, 115674 (2020).
6. Tian, J. et al. Applying ultraviolet/persulfate (UV/PS) preoxidation for controlling ultrafiltration membrane fouling by natural organic matter (NOM) in surface water. *Water Res.* **132**, 190–199 (2018).
7. Xu, J., Chang, C.-Y. & Gao, C. Performance of a ceramic ultrafiltration membrane system in pretreatment to seawater desalination. *Sep. Purif. Technol.* **75**, 165–173 (2010).
8. Chen, M. et al. Spinel-based ceramic membranes coupling solid sludge recycling with oily wastewater treatment. *Water Res.* **169**, 115180 (2020).
9. Mo, J., Li, X. & Yang, Z. Dissecting the structure-property relationship of ceramic membrane with asymmetric multilayer structures for maximizing permselectivity. *Water Res.* **220**, 118658 (2022).
10. Zou, D. et al. Co-sintering of ceramic ultrafiltration membrane with gradient pore structures for separation of dye/salt wastewater. *Sep. Purif. Technol.* **302**, 122030 (2022).
11. Fan, S. et al. Rational design of dynamic fibre membrane for sustainable biofouling control. *Nat. Water* **2**, 161–171 (2024).
12. Han, S. et al. Microporous organic nanotube assisted design of high performance nanofiltration membranes. *Nat. Commun.* **13**, 7954 (2022).
13. Chen, L. et al. In situ visualization of membrane fouling evolution during ultrafiltration using label-free hyperspectral light sheet fluorescence imaging. *Environ. Sci. Technol.* **57**, 4533–4542 (2023).
14. Wang, Y., Ma, B., Ulbricht, M., Dong, Y. & Zhao, X. Progress in alumina ceramic membranes for water purification: Status and prospects. *Water Res.* **226**, 119173 (2022).
15. Murzyn, K. et al. Breaking Abbe's diffraction limit with harmonic deactivation microscopy. *Sci. Adv.* **10**, eadp3056 (2024).
16. Wu, S. et al. Three-dimensional analysis of the natural-organic-matter distribution in the cake layer to precisely reveal ultrafiltration fouling mechanisms. *Environ. Sci. Technol.* **55**, 5442–5452 (2021).
17. Li, D. et al. Nodular networks in hydrated polyamide desalination membranes enhance water transport. *Sci. Adv.* **11**, eadt3324 (2025).
18. Kalutantirige, F. C. et al. Beyond nothingness in the formation and functional relevance of voids in polymer films. *Nat. Commun.* **15**, 2852 (2024).
19. Culp, T. E. et al. Nanoscale control of internal inhomogeneity enhances water transport in desalination membranes. *Science* **371**, 72–75 (2021).
20. Collinson, L. M. et al. Volume EM: a quiet revolution takes shape. *Nat. Methods* **20**, 777–782 (2023).
21. Heinrich, L. et al. Whole-cell organelle segmentation in volume electron microscopy. *Nature* **599**, 141–146 (2021).
22. Azad, R. et al. Medical image segmentation review: the success of U-net. *IEEE Trans. Pattern Anal. Mach. Intell.* **46**, 10076–10095 (2024).
23. Gostick, J. et al. OpenPNM: A pore network modeling package. *Comput. Sci. Eng.* **18**, 60–74 (2016).
24. Daniel, M.-C. & Astruc, D. Gold nanoparticles: Assembly, supramolecular chemistry, quantum-size-related properties, and applications toward biology, catalysis, and nanotechnology. *Chem. Rev.* **104**, 293–346 (2004).
25. Garcia-Ivars, J. et al. Nanofiltration as tertiary treatment method for removing trace pharmaceutically active compounds in wastewater from wastewater treatment plants. *Water Res.* **125**, 360–373 (2017).
26. Wu, H. et al. Treatment of oily wastewaters by highly porous whisker-constructed ceramic membranes: Separation performance and fouling models. *Water Res.* **211**, 118042 (2022).
27. Zhang, Y., Wang, J., Gao, F., Chen, Y. & Zhang, H. A comparison study: The different impacts of sodium hypochlorite on PVDF and PSF ultrafiltration (UF) membranes. *Water Res.* **109**, 227–236 (2017).
28. Chang, Z. H., Sum, J. Y., Leow, L. H. T., Teow, Y. H. & Yeap, S. P. Interfacial interaction energy and pore blockage analysis of protein fouling in ultrafiltration-like forward osmosis. *Appl. Surf. Sci.* **693**, 162787 (2025).
29. Yin, J., Tang, H., Xu, Z. & Li, N. Enhanced mechanical strength and performance of sulfonated polysulfone/tröger's base polymer blend ultrafiltration membrane. *J. Membr. Sci.* **625**, 119138 (2021).
30. Qin, G., Liu, Y., Rietveld, L. C. & Heijman, S. G. J. Oilfield-produced water treatment with SiC-coated alumina membranes. *Sep. Purif. Technol.* **362**, 131841 (2025).
31. Su, T. J. et al. Identification of the location of protein fouling on ceramic membranes under dynamic filtration conditions. *J. Membr. Sci.* **163**, 265–275 (1999).
32. Fan, Y., Yi, J., Zhang, Y. & Yokoyama, W. Fabrication of curcumin-loaded bovine serum albumin (BSA)-dextran nanoparticles and the cellular antioxidant activity. *Food Chem.* **239**, 1210–1218 (2018).
33. De Rosis, A. & Coreixas, C. Multiphysics flow simulations using D3Q19 lattice boltzmann methods based on central moments. *Phys. Fluids* **32**, 117101 (2020).

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Conceptualization: D.S., T.W., R.L., Y.Z. and X.Z. Methodology: D.S., T.W. and R.L. Investigation: D.S. Visualization: D.S. Supervision: R.L. and X.Z. Writing—original draft: D.S. Writing—review & editing: D.S., T.W., R.L., X.Z., M.S., H.L. and J.Q.

Competing interests

The authors declare no competing interests.

Additional information

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